Toward a four-dimensional Yang–Mills construction

Notebook 16: local nonlinear responses and the classical coarse action

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Abstract

Starting from the finite-regulator small-field minimizer of Notebook 15, we prove exponential locality of its response to coarse data at small nonconstant backgrounds. The bounds are uniform in refinement and periodic volume, retain the same physical Hölder exponent, and allow one weighted direction at every analytic derivative order. The extra input is a local physical-transpose derivative estimate for the nonlinear constraint. An exact envelope identity then identifies the derivative of the minimized classical action with the coarse multiplier. This supplies a local nonlinear coarse Hessian, its small perturbation from the flat Maxwell precision, and a classical gauge identity. A separate exact Gaussian trace calculation shows that a fixed physical smooth ball has vanishing probability when the coupling decreases more slowly than the square of the mesh spacing. This identifies a concrete need for rescaled fluctuation domains. Notebook 17 supplies the resulting fixed-step Gaussian domain and joint tails; Notebook 18 supplies an overlapping-interaction integration theorem under explicit local coefficient hypotheses. The complete model-specific integration and iteration remain open.

Status. Internally audited research proofs, with no external certification or novelty claim. The prerequisite finite-regulator construction and its ordinary-constraint transfer are given in Notebook 15. The source provenance remains the deposited transcriptions at commit `e392ef45013a581664bab76b90867170b30fe10e`. No new original-paper verification is asserted here.

1 The precise target

Work with $SU(N_c)$ on compatible finite periodic lattices, fine spacing $\eta = L^{-j}$ for fixed odd $L \geq 3$, coarse spacing one, and fixed $0 < \alpha < 1$. The background coarse potential B is arbitrary in a sufficiently small global supremum ball; it need not be constant or decay at infinity. The selected fine potential $A_*(B)$ uses the corrected constraint and fixed flat gauge of Notebook 15. All derivatives of this chosen representative below refer to that selection.

The ordinary-feasible representative from Notebook 15 is gauge equivalent and has a separately controlled response through its explicit gauge formula. Its leading extension generally differs from H, while its minimized action value is exactly the same. Keeping these coordinates distinct gives local response theorems for both specified representatives and the ordinary-gauge interpretation of the classical action.

The main analytic estimate puts an exponential weight on one coarse direction while keeping the other directions and the background unweighted. It yields decay away from a localized insertion and uniform higher derivatives. A fixed-exponent connected-tree bound involving all insertion sites at once is a stronger statement and is not inferred from this estimate.

2 Exponential locality of nonlinear minimizer responses

Fix $0 < \alpha < 1$, an odd blocking ratio $L \geq 3$, refinement $M = L^j$, fine spacing $\eta = M^{-1}$, and final spacing one. The primary statements are on finite compatible periodic regulators, with constants independent of refinement and volume. The coarse background may be nonconstant, but it lies in a fixed small complex bounded-field ball. We study the corrected-constraint minimizer $A(B)$ in its fixed flat gauge. A gauge-equivalent representative chosen by a different construction does not acquire these locality estimates merely from gauge equivalence.

2.1 Weighted spaces and the fixed flat inputs

Write

$$\mathcal{F}(A) = QA + \mathcal{R}(A), \qquad D_A = D\mathcal{R}(A), \qquad \mathcal{X} = \ker Q \cap \ker(RD^T).$$

Here R is the fixed scalar gauge projection and $\mathcal{R}$ is the nonlinear averaging remainder. All transposes use physical measure. Let $\mathcal{N}_\alpha$ be the complete potential norm consisting of the supremum, the suprema of all physical first differences, and their local α-Hölder seminorms. Fix unit-cell collars of bounded physical size.

Choose a sufficiently small fixed $\kappa > 0$ allowed by the flat kernel and constrained Schauder estimates. An admissible positive cell weight satisfies

$$\frac{w(n)}{w(m)} \leq e^{\kappa d(n,m)}. \tag{1}$$

Weighted norms take the supremum of $w(n)$ times the appropriate local norm on the collar of cell n. Fine derivatives are taken on complete fields before these norms are evaluated. No sharp cell cutoff or derivative of the weight is inserted. Periodic distance is used on a torus, and fixed collar enlargements only change constants.

On the infinite lattice a decaying admissible weight can allow unbounded weighted directions. Literal Fréchet responses of the original bounded-field map are first taken on globally bounded directions with finite weighted norm. The linearized formulas below then define bounded extensions to the full weighted spaces. Higher-response symmetry is initially understood on bounded directions. The localized corollaries use growing weights $w_S \geq 1$ and require no such extension.

The proved flat operator estimates, uniform for all these weights, are

$$H : \ell^\infty_{c,w} \longrightarrow \mathcal{N}_{\alpha,w}, \qquad \mathcal{C} : \ell^\infty_{f,w} \longrightarrow \mathcal{N}_{\alpha,w}, \tag{2}$$

$$\mathcal{C}S^T_{\mathrm{jet}} : C^\alpha_{f,w} \longrightarrow \mathcal{N}_{\alpha,w}, \qquad q : \ell^\infty_{c,w} \longrightarrow \ell^\infty_{c,w}, \tag{3}$$

$$H^T : \ell^\infty_{f,w} \longrightarrow \ell^\infty_{c,w}, \qquad (S_{\mathrm{jet}}H)^T : \ell^\infty_{f,w} \longrightarrow \ell^\infty_{c,w}. \tag{4}$$

The symbol c denotes coarse fields and f fine fields. The exponent in (3) is unchanged; the separately proved constrained biharmonic cancellation is needed for this. The transpose bounds in (4) use the physical volume one of a fine cell and summable derivative-extension kernels.

Write the nonlinear Wilson force as $N(A) = S^T_{\mathrm{jet}} f_\eta(S_{\mathrm{jet}}A)$. The local analytic coefficient and the complete-field Hölder product estimate give

$$\|D[f_\eta(S_{\mathrm{jet}}A)]V\|_{C^\alpha_w} \leq C \|A\|_{\mathcal{N}_\alpha} \|V\|_{\mathcal{N}_{\alpha,w}}. \tag{5}$$

Thus $\mathcal{C}N'(A)$ and $H^T N'(A)$ have the corresponding weighted norms bounded by $C_\kappa \|A\|_{\mathcal{N}_\alpha}$. The second operator is evaluated using $(S_{\mathrm{jet}}H)^T$; no rough nonlinear coefficient is differentiated.

2.2 A local estimate for the differentiated transpose

The full corrected average has a physical dependency radius R_* independent of j. One may use the deposited two-block property, or sum the one-step radii using $\sum_{t<j} h_t \le (L-1)^{-1}$. Increase R_* by a fixed cell collar margin. The established absolute physical row estimates are

$$\|D_A\|_{\infty\to\infty} \le c\,\|A\|_\infty\,, \qquad \left\|D_A^T\right\|_{\infty\to\infty} \le c_*\,\|A\|_\infty\,. \tag{6}$$

These inequalities hold in the absolute row norm as well. For the second inequality the one-step physical column factor is L^d, giving $N_L K_0 L^{d-1} h\,\|A\|_\infty$ before summing the spacings. It is not a comparison with the number of fine variables.

Finite dependency immediately yields

$$\begin{aligned} \|D_A V\|_{\infty,w} &\le c e^{\kappa R_*}\,\|A\|_\infty\,\|V\|_{\infty,w}\,,\\ \left\|D_A^T m\right\|_{\infty,w} &\le c_* e^{\kappa R_*}\,\|A\|_\infty\,\|m\|_{\infty,w}\,. \end{aligned} \tag{7}$$

The key additional estimate places the weight on a background variation while leaving the coarse multiplier unweighted:

$$\boxed{\left\|(D_A^T)'[V]m\right\|_{\infty,w} \le C_* e^{2\kappa R_*}\,\|V\|_{\infty,w}\,\|m\|_\infty\,.} \tag{8}$$

The constant is uniform on a fixed smaller complex background ball.

Proof of (8). Fix an output fine bond b. Its row in D_A^T reads only coarse bonds within R_* of b. Each associated coefficient depends on A only in the dependency set of that coarse bond, hence within $2R_*$ of b. The map from the bounded background ball into this physical absolute row space is analytic and uniformly bounded by (6). Cauchy on a fixed larger ball bounds its derivative by $C_*\,\|V\|_\infty$.

For this one row restrict V to the union of those dependency sets. The restriction does not change the row derivative, and its supremum is at most $e^{2\kappa R_*} w(b)^{-1}\,\|V\|_{\infty,w}$, with the fixed collar convention absorbed in R_*. This restriction is used only in a bounded-value coefficient map, where no field derivative is taken. Sum the resulting physical row against bounded m, multiply by $w(b)$, and take the supremum over b. The same argument works with lifted multiplicities on small tori. □

The doubled dependency radius in this proof is essential. Cauchy is applied to the uniformly bounded physical row, not separately to every fine coefficient followed by an uncontrolled count. Higher coefficient derivatives admit the analogous estimate with one weighted direction and the remaining directions bounded, with factorial constants from the analytic margin. The earlier two-Hilbert-direction Hessian incidence estimate is consistent with this bound but does not replace its proof.

2.3 Weighted variation of the chart and multiplier

The exact chart and multiplier are

$$\begin{aligned} A &= \xi + H[B - \mathcal{R}(A)],\\ m &= (I + H^T D_A^T)^{-1}\{q[B - \mathcal{R}(A)] + H^T N(A)\}. \end{aligned} \tag{9}$$

Let r bound $\|A\|_{\mathcal{N}_\alpha}$ and b bound $\|B\|_\infty$. The established unweighted estimate is

$$\|m\|_\infty \le C(b + r^2). \tag{10}$$

Neither A nor m is required to have a finite growing-weight norm. Only their variations will carry a weight.

Shrink the common background radius once more, depending on κ, so that uniformly in admissible w,

$$\|HD_A\|_{\mathcal{N}_{\alpha,w}\to\mathcal{N}_{\alpha,w}} < \frac{1}{2}, \qquad \left\|H^T D_A^T\right\|_{\infty,w\to\infty,w} < \frac{1}{2}. \tag{11}$$

These inequalities follow from (2), (4), and (7). Their radii are independent of source location, refinement, and volume.

For $z = \delta\xi$, $b_1 = \delta B$, and $v = \delta A$, exact differentiation gives

$$v = P_A(z + Hb_1), \qquad P_A = (I + HD_A)^{-1}, \qquad \|v\|_{\mathcal{N}_{\alpha,w}} \le 2(\|z\|_{\mathcal{N}_{\alpha,w}} + h_\kappa \|b_1\|_{\infty,w}). \tag{12}$$

Differentiate the defining equation for m in (9). This preserves the noncommuting order and gives

$$\boxed{\delta m = (I + H^T D_A^T)^{-1}\{q(b_1 - D_A v) + H^T N'(A)v - H^T (D_A^T)'[v]m\}.} \tag{13}$$

Using (5), (7), (8), and (10),

$$\|\delta m\|_{\infty,w} \le C_\kappa(\|b_1\|_{\infty,w} + (r + b + r^2)\|v\|_{\mathcal{N}_{\alpha,w}}). \tag{14}$$

In particular the last term of (13) needs the mixed transpose estimate; no weighted norm of the global background multiplier has been assumed.

2.4 The weighted response inverse

The selected minimizer solves

$$\xi = \Phi_B(\xi) = -\mathcal{C}N(A) + \mathcal{C}D_A^T m. \tag{15}$$

Its variation is exactly

$$\delta\Phi = -\mathcal{C}N'(A)v + \mathcal{C}\{(D_A^T)'[v]m + D_A^T\delta m\}. \tag{16}$$

The preceding estimates imply, for $r < 1$,

$$\|\delta\Phi\|_{\mathcal{N}_{\alpha,w}} \le C_\kappa((r + b)\|v\|_{\mathcal{N}_{\alpha,w}} + r\|b_1\|_{\infty,w}). \tag{17}$$

For $b_1 = 0$, combine this with (12). Choose the chart and coarse background radii so that $2C_\kappa(r + b) < 1/2$. Then

$$\|D_\xi\Phi_B\|_{\mathcal{N}_{\alpha,w}\cap\mathcal{X}\to\mathcal{N}_{\alpha,w}\cap\mathcal{X}} < \frac{1}{2}. \tag{18}$$

Every term lies in $\mathcal{X}$ because it contains $\mathcal{C}$. The constrained space is closed on each finite regulator. The estimate does not use a bounded-field norm for the separate gauge projection.

The derivative of the minimizer is therefore obtained by the weighted Neumann inverse

$$z = (I - D_\xi\Phi_B)^{-1}D_B\Phi_B[b_1], \qquad D_B A(B)[b_1] = P_A(z + Hb_1). \tag{19}$$

At the solution $\|A(B)\|_{\mathcal{N}_\alpha} \le C\|B\|_\infty$. Equations (17) and (19) show that z is $O(\|B\|_\infty \|b_1\|_{\infty,w})$. Also $P_A - I = -P_A HD_A$ is $O(\|B\|_\infty)$ in the weighted norm. Consequently

$$\boxed{\begin{aligned}\|D_B A(B)[b_1]\|_{\mathcal{N}_{\alpha,w}} &\le C_\kappa \|b_1\|_{\infty,w}\,,\\ \|(D_B A(B) - H)[b_1]\|_{\mathcal{N}_{\alpha,w}} &\le C_\kappa \|B\|_\infty \|b_1\|_{\infty,w}\,.\end{aligned}} \tag{20}$$

This is a response estimate at arbitrary small nonconstant backgrounds derived using fixed flat operators and nonlinear coefficients. No nonflat source Schauder theorem is assumed.

2.5 Coarse multiplier responses

Now set $m(B) = m(A(B), B)$ and $v = D_B A(B)[b_1]$ in (13). It immediately gives

$$\|D_B m(B)[b_1]\|_{\infty,w} \le C_\kappa \|b_1\|_{\infty,w}. \tag{21}$$

More explicitly, putting $M_A = I + H^T D_A^T$ and subtracting qb_1 gives the exact expression

$$\begin{aligned}(D_B m(B) - q)b_1 = (M_A^{-1} - I)qb_1 \\ + M_A^{-1}\{-qD_A v + H^T N'(A)v - H^T (D_A^T)'[v]m\}.\end{aligned} \tag{22}$$

At the solution the inverse difference, D_A, and $N'(A)$ are $O(\|B\|_\infty)$ in their respective weighted spaces. Although $(D_A^T)'$ need not vanish at zero, its multiplier has norm $\|m\|_\infty = O(\|B\|_\infty)$, so (8) controls that term too. Thus

$$\boxed{\|(D_B m(B) - q)b_1\|_{\infty,w} \le C_\kappa \|B\|_\infty \|b_1\|_{\infty,w}.} \tag{23}$$

The multiplier is holomorphic as a global bounded-field map, and its derivative is holomorphic in these weighted operator spaces by the displayed formulas and the uniformly convergent weighted Neumann series.

2.6 Localized changes and all-order bounds with one weight

If b_1 is supported in coarse cells S, choose $w_S(n) = e^{\kappa d(n,S)}$. This is admissible. Equation (20) gives

$$\|D_B A(B)[b_1]\|_{\mathcal{N}_\alpha(C_n^+)} \le C_\kappa e^{-\kappa d(n,S)} \|b_1\|_\infty. \tag{24}$$

Equations (21) and (23) give the analogous coarse multiplier estimates. For two small backgrounds whose connecting coarse line segment stays inside the common ball, integrate these derivative bounds to obtain

$$\|A(B') - A(B)\|_{\mathcal{N}_{\alpha,w}} + \|m(B') - m(B)\|_{\infty,w} \le C_\kappa \|B' - B\|_{\infty,w}. \tag{25}$$

The individual backgrounds may be globally nondecaying. On the full lattice truncated admissible weights and passage to their limit justify the same statement without assuming the difference in advance belongs to the growing-weight space.

On a smaller common complex background ball, $B \mapsto D_B A(B)$ and $B \mapsto D_B m(B)$ are holomorphic into the weighted operator spaces. On finite regulators this is finite-dimensional holomorphy with the uniform bounds just proved. Equivalently it follows from the local coefficient analyticity, the Hölder product estimates, and the weighted inverse formulas, which also apply to complete infinite fields. Background directions in this assertion are unweighted bounded coarse fields. If the singled-out weighted direction is unbounded on the infinite lattice, the operator-valued formulas specify the linearized extension just described, rather than a Fréchet derivative in an inadmissible background direction.

Let $\delta > 0$ be the margin to a larger complex background ball where the uniform weighted derivative bounds hold. Operator-valued multivariable Cauchy gives, for every $n \ge 1$,

$$\begin{aligned}\boxed{\|D_B^n A(B)[b_1, \dots, b_n]\|_{\mathcal{N}_{\alpha,w}} + \|D_B^n m(B)[b_1, \dots, b_n]\|_{\infty,w}} \\ \le C_\kappa (n-1)! \left(\frac{e}{\delta}\right)^{n-1} \|b_1\|_{\infty,w} \prod_{i=2}^{n} \|b_i\|_\infty.\end{aligned} \tag{26}$$

For $n = 1$ this is the proved first-response estimate. For $n > 1$ use polyradii approaching $\delta/(n-1)$ and $(n-1)^{n-1} \le e^{n-1}(n-1)!$. Symmetry permits any single direction to carry the weight.

If all directions are supported in sets S_i, applying (26) separately with each direction weighted gives decay $\exp[-\kappa \max_i d(n, S_i)]$ at an output cell n. This is a precise one-weight conclusion. It is not a fixed-exponent tree-length or ordered-path estimate among all insertion sites. Such stronger multivertex bounds require an additional branched expansion or multiweight argument.

2.7 Scope of the response theorem

The exponent κ is fixed once, and the allowed background ball is reduced by a fixed amount to satisfy the weighted inverse inequalities. Its radius does not depend on refinement, volume, the insertion location, or the particular admissible weight. The theorem does not assert that the largest unweighted existence radius already has this property.

These results concern the selected corrected-constraint minimizer in the fixed flat gauge and its coarse multiplier. They do not identify its derivatives with a moving-gauge source minimizer, establish mixed-history estimates, or prove an interacting continuum limit. All action-minimum statements remain finite regulator statements; bounded infinite fields need not have finite total action.

3 Locality of the chosen ordinary-constraint representative

Gauge equivalence by itself does not preserve locality of a chosen potential representative. For the explicit gauge used in the ordinary constraint transfer, however, the required locality can be proved. Write $A_*(B)$ for the corrected fixed-slice minimizer and retain the cumulative endpoint gauge b_j and the fixed bump extension E_{pt} from that transfer. Set

$$\omega(A) = \frac{1}{i} \log b_j(e^{i\eta A})^{-1}, \qquad \lambda(A) = E_{\mathrm{pt}}\omega(A), \qquad T(A) = \mathscr{A}_\eta(A, \lambda(A)), \qquad A_{\mathrm{ord}}(B) = T(A_*(B)). \tag{27}$$

All bases are small in the unweighted global norms. Weights are placed on variations only. Fix $0 < \alpha < 1$ and one sufficiently small positive spatial exponent κ.

3.1 The derivative of the gauge transfer is weighted bounded

The cumulative gauge is uniformly holomorphic in the global potential supremum norm, has $b_j - I = O(\|A\|_\infty)$, and has a fixed final physical dependency radius R_b. The latter follows from the endpoint product and the summable physical reading radii of the corrected trajectory. The same properties hold for ω, with $\omega(0) = 0$. For an admissible weight $w(n)/w(m) \le e^{\kappa d(n,m)}$, local-row Cauchy gives

$$\|D\omega(A)V\|_{\infty,w} \le Ce^{\kappa R_b} \|V\|_{\infty,w}. \tag{28}$$

Indeed, for one coarse output row, restrict the direction to that row's dependency neighborhood. Its unweighted supremum is bounded by the right side divided by the output weight. No derivative is taken of this restricted direction: ω consumes bounded potential values. Cauchy is applied to the whole bounded row, not coefficientwise followed by a fine-dimensional count. The same argument proves operator-valued holomorphy in the weighted norm on a smaller unweighted background ball.

The fixed bump interpolation satisfies

$$\|E_{\mathrm{pt}}c\|_{C^{2,\alpha}_w} \le C_\kappa \|c\|_{\infty,w}. \tag{29}$$

Each collar reads a fixed number of nearby coarse coefficients. Bounded smooth bump derivatives through order three, weight comparison on that neighborhood, and sampling give this estimate, uniformly in refinement and compatible periodic volume. Consequently

$$\|D\lambda(A)V\|_{C^{2,\alpha}_w} \le C_\kappa \|V\|_{\infty,w}, \qquad \|\lambda(A)\|_{C^{2,\alpha}} \le C \|A\|_\infty. \tag{30}$$

The base estimate in the second inequality is unweighted.

For completeness, the fine gauge action is

$$\mathscr{A}_\eta(A,\lambda)(e) = \frac{1}{i\eta}\log(e^{i\lambda(e_-)}e^{i\eta A(e)}e^{-i\lambda(e_+)}).$$

Writing $\ell = \lambda(e_-)$ and $v = (\lambda(e_+) - \lambda(e_-))/\eta$, its group argument is $e^{i\eta\,\mathrm{Ad}_{e^{i\ell}}A(e)}e^{i\ell}e^{-i(\ell+\eta v)}$. It equals identity at $\eta = 0$, so its logarithm divided by η has a uniform analytic jet extension there. The derivative is therefore a finite sum of products of analytic coefficients of the base jet $(A, \lambda, D\lambda)$ and the variation jet $(V, \mu, D\mu)$. The base coefficients have bounded unweighted $C^{1,\alpha}$ norms. The discrete product estimate with one weighted factor gives

$$\|D\mathscr{A}_\eta(A,\lambda)[V,\mu]\|_{N_{\alpha,w}} \le C_\kappa(\|V\|_{N_{\alpha,w}} + \|\mu\|_{C^{2,\alpha}_w}). \tag{31}$$

A first output difference uses at most two scalar differences. The same jet estimate controls its Hölder member; there is no inverse-mesh loss or weighted-smallness assumption on the base. Combining (30) and (31) proves

$$\|DT(A)V\|_{N_{\alpha,w}} \le C_\kappa \|V\|_{N_{\alpha,w}}, \tag{32}$$

with holomorphic dependence in this weighted operator norm.

3.2 Responses and the new leading extension

The corrected minimizer's weighted response theorem and the chain rule now give

$$D_B A_{\mathrm{ord}}(B) = DT(A_*(B))D_B A_*(B), \qquad \|D_B A_{\mathrm{ord}}(B)b\|_{N_{\alpha,w}} \le C_\kappa \|b\|_{\infty,w}. \tag{33}$$

Its leading extension is

$$H_{\mathrm{ord}} = DT(0)H, \tag{34}$$

and can differ from H. To check the sign explicitly, let $\beta(A) = \log b_j(e^{i\eta A})/i$ and $B_{\mathrm{cum}} = D\beta(0)$. On the common branch $\omega = -\beta$, and the gauge action has derivative $V - D\mu$ at zero. Hence

$$DT(0) = I + DE_{\mathrm{pt}}B_{\mathrm{cum}}, \qquad H_{\mathrm{ord}} = H + DE_{\mathrm{pt}}B_{\mathrm{cum}}H. \tag{35}$$

Thus $dH_{\mathrm{ord}} = dH$. The exact transfer identity $\mathcal{F}_{\mathrm{ord}}(T(A)) = \mathcal{F}_{\mathrm{corr}}(A)$ also gives $Q_{\mathrm{ord}}DT(0) = Q$ and $Q_{\mathrm{ord}}H_{\mathrm{ord}} = I$, where Q_{ord} is the derivative of the ordinary homogeneous constraint at zero. Here $\mathcal{F}_{\mathrm{ord}}(A) = \log M^{(j)}(e^{i\eta A})/i$ and $\mathcal{F}_{\mathrm{corr}} = \mathcal{F}$; the complete norm N_α is the same as $\mathcal{N}_\alpha$ used above.

The response family in (33) is holomorphic on a common unweighted complex coarse ball with a uniform weighted operator norm. Operator Cauchy on a smaller ball yields

$$\|(D_B A_{\mathrm{ord}}(B) - H_{\mathrm{ord}})b\|_{N_{\alpha,w}} \le C_\kappa \|B\|_\infty \|b\|_{\infty,w}. \tag{36}$$

If δ is a fixed analytic margin, then for every $n \ge 1$,

$$\|D_B^n A_{\mathrm{ord}}(B)[b_1,\dots,b_n]\|_{N_{\alpha,w}} \le C_\kappa(n-1)!\left(\frac{e}{\delta}\right)^{n-1}\|b_1\|_{\infty,w}\prod_{i=2}^{n}\|b_i\|_\infty. \tag{37}$$

For $n > 1$ apply multivariable Cauchy to the first-response operator with $n-1$ unweighted directions and use $(n-1)^{n-1} \le e^{n-1}(n-1)!$. The case $n = 1$ is (33). Symmetry permits any selected bounded direction to carry the weight.

For a direction supported in S, choose $w_S(n) = e^{\kappa d(n,S)}$. Equations (33) and (37) give exponential decay away from S. If all directions are supported in sets S_i, applying the bound separately to each

weighted direction gives the factor $e^{-\kappa \max_i d(n,S_i)}$. No tree-length estimate is inferred. Integration along a straight coarse segment gives

$$\left\|A_{\mathrm{ord}}(B') - A_{\mathrm{ord}}(B)\right\|_{N_{\alpha,w}} \le C_\kappa \left\|B' - B\right\|_{\infty,w}. \tag{38}$$

Thus finite changes supported in S have the same spatial decay. For complete infinite fields use bounded truncated weights and pass to the limit. For weights that decay to zero, literal Fréchet responses are first read on bounded directions with finite weighted norm; the formulas define their bounded extension to the full weighted space. The support weights satisfy $w_S \ge 1$ and require no such qualification.

This proves locality of the particular ordinary representative defined in (27); the field-dependent gauge has been estimated rather than treated as harmless by gauge equivalence. It does not assert locality of every gauge-equivalent representative, identify the separately moving source gauge, or cover mixed retained-history geometries. The action-minimum statement remains the finite homogeneous small-global-field theorem proved earlier.

4 The classical coarse action and its gauge identity

We remain on finite compatible periodic regulators, with fine spacing $\eta = L^{-j}$ and final coarse spacing one. The action $\mathcal{S}_\eta(U)$ is the coupling-free Wilson action in the normalization of Notebook 15. Set $\mathcal{W}_\eta(A) = \mathcal{S}_\eta(e^{i\eta A})$. All coarse pairings below use the normalized color trace and the physical coarse measure, which is one at this scale. Transposes are bilinear after complexification. The constants are independent of refinement and periodic volume.

Let $A_*(B)$ be the corrected-constraint minimizer in the fixed flat gauge slice, and put $U_*(B) = e^{i\eta A_*(B)}$. Its ordinary-feasible representative $U_{\mathrm{ord}}(B)$ is related to $U_*(B)$ by the genuine gauge transformation constructed in Notebook 15. Define

$$V(B) = \mathcal{W}_\eta(A_*(B)) = \mathcal{S}_\eta(U_*(B)) = \mathcal{S}_\eta(U_{\mathrm{ord}}(B)). \tag{39}$$

These are equal holomorphic functions on a uniform complex coarse supremum ball. Indeed the analytic Wilson density, written with $P + P^{-1}$, is invariant under complex matrix gauge conjugation as well. Only the real fields have the minimum interpretation. The scalar total action scales with volume; the uniform estimates below state that dependence.

4.1 The multiplier is the exact derivative of the action

Write $\mathcal{F}(A) = QA + \mathcal{R}(A)$ and use the exact chart and flat-slice space

$$A = \xi + H[B - \mathcal{R}(A)], \qquad \xi \in \mathcal{X} = \ker Q \cap \ker(R_0 D^T).$$

Here R_0 is the fixed scalar gauge projection, distinct from the nonlinear remainder $\mathcal{R}$ and denoted R in the response module. Put

$$T_A = (I + HD\mathcal{R}(A))^{-1}, \qquad H_A = T_A H. \tag{40}$$

Differentiating the analytic minimizer in a coarse direction c gives the exact formula

$$D_B A_*(B)[c] = H_A c + T_A D_B \xi(B)[c], \qquad A = A_*(B). \tag{41}$$

The second summand is a slice tangent because $D_B\xi(B)[c] \in \mathcal{X}$.

Proposition 4.1 (Classical envelope identity)**.** *Let $E(A) = D\mathcal{W}_\eta(A)$ denote the Euler field under the physical fine pairing. The multiplier from the exact stationarity construction satisfies*

$$DV(B)[c] = \langle m(B), c\rangle_1, \qquad m(B) = H_A^T E(A_*(B)). \tag{42}$$

Equivalently,

$$\begin{aligned} m(B) = (I + H^T D\mathcal{R}(A_*)^T)^{-1} \\ \cdot \{q[B - \mathcal{R}(A_*)] + H^T N(A_*)\}, \end{aligned} \tag{43}$$

where $q = q_M$ is the physical flat Maxwell precision and $E(A) = KA + N(A)$ is the ordinary-coordinate force.

Proof. Pair (41) with $E(A_*)$. Slice stationarity makes $\langle E(A_*), T_A D_B \xi(B)[c]\rangle_\eta = 0$. The remaining term is exactly (42). This argument needs only slice stationarity, not a separate full-constraint multiplier theorem or gauge coverage.

The transpose push-through identity gives $H_A^T = (I + H^T D\mathcal{R}(A_*)^T)^{-1} H^T$. The flat identities $H^T K\xi = 0$ and $H^T K H = q$ imply $H^T K A_* = q[B - \mathcal{R}(A_*)]$, proving (43). The force term is evaluated as $H^T N(A_*) = (S_{\text{jet}} H)^T f_\eta(S_{\text{jet}} A_*)$; no rough force coefficient is differentiated. All identities extend bilinearly to the complex ball. □

Put $b = \|B\|_\infty$. The uniform estimates $A_* = HB + O(b^2)$, $\mathcal{R}(A_*) = O(b^2)$, $H^T N(A_*) = O(b^2)$ and $\left\|D\mathcal{R}(A_*)^T\right\|_{\infty\to\infty} = O(b)$ in (43) prove

$$\|m(B) - qB\|_\infty \le Cb^2, \qquad m(0) = 0, \quad Dm(0) = q. \tag{44}$$

The multiplier is holomorphic on a uniform coarse ball. In particular $D^2V(0) = q$. Its unweighted estimate $\|Dm(B) - q\|_{\infty\to\infty} \le Cb$ also follows directly by Cauchy: for $b > 0$ use a direction circle of radius b inside a smaller common ball, and apply (44) at points of norm at most $2b$. At $b = 0$ use the displayed Taylor coefficient.

Let N_b denote the number of independent positive coarse bonds. Since $V(0) = 0$, radial integration gives

$$V(B) = \frac{1}{2}\langle B, qB\rangle_1 + \int_0^1 \langle m(tB) - tqB, B\rangle_1 \, dt. \tag{45}$$

The fixed normalized color trace bound and (44) therefore yield

$$\boxed{\left|V(B) - \frac{1}{2}\langle B, qB\rangle_1\right| \le CN_b \|B\|_\infty^3 .} \tag{46}$$

This holds on the complex ball. It is a volume-uniform density estimate, not a volume-independent total-action bound.

4.2 Local coarse gauge invariance

For a real coarse site field ω put $g = e^{i\omega}$, $D = e^{iB}$ and

$$D^g(c) = g(c_-)D(c)g(c_+)^{-1}, \qquad B^g = \frac{1}{i}\log D^g. \tag{47}$$

All logarithms use the common small branch. At coarse spacing one the local analytic multiplication gives a constant $C_c \ge 1$ such that

$$\|B^g\|_\infty \le C_c(b + w), \qquad w = \|\omega\|_\infty . \tag{48}$$

No additional regularity is required of ω: coarse first differences are bounded by $2w$.

Let b_0, r_0 be the ordinary-minimum radii of Notebook 15, and write $\|A_{\mathrm{ord}}(B)\|_{N_\alpha} \le C_A b$. The smooth point-value extension there satisfies

$$(E_{\mathrm{pt}}\omega)|_{\mathrm{coarse}} = \omega, \qquad \|E_{\mathrm{pt}}\omega\|_{C^{2,\alpha}} \le C_E w. \tag{49}$$

Set $h = e^{iE_{\mathrm{pt}}\omega}$, so $h|_{\mathrm{coarse}} = g$. The fine gauge-action bound, valid for both h and h^{-1}, is

$$\left\|A^h\right\|_{N_\alpha} \le C_G(\|A\|_{N_\alpha} + C_E w). \tag{50}$$

Choose one $\delta > 0$ so that all logarithms and gauge actions stay in their established domains and

$$C_c\delta < b_0/2, \qquad C_G(C_A C_c + C_E)\delta < r_0/2. \tag{51}$$

For real $b + w < \delta$, ordinary gauge covariance makes $U_{\mathrm{ord}}(B)^h$ feasible for coarse datum B^g. Its potential norm is below r_0, so minimality at B^g and exact gauge invariance give

$$V(B^g) \le \mathcal{S}_\eta(U_{\mathrm{ord}}(B)^h) = V(B).$$

For the reverse comparison use $U_{\mathrm{ord}}(B^g)^{h^{-1}}$. Its ordinary coarse average is $(D^g)^{g^{-1}} = D$, and its norm is bounded by

$$C_G\{C_A C_c(b + w) + C_E w\} < r_0.$$

It is therefore an admissible competitor at B, proving $V(B) \le V(B^g)$. We have proved

$$\boxed{V(B^g) = V(B) \qquad (b + w < \delta).} \tag{52}$$

Both comparisons lie strictly inside the same uniform minimum radius. The reverse one uses the separately controlled minimizer at B^g, not an assumed inverse of the first comparison map.

The two sides of (52) are holomorphic in (B, ω) on a uniform smaller complex product ball. Their equality on an open set of the full real form implies equality on that connected product ball by the finite-dimensional holomorphic identity theorem, applied coordinate by coordinate. For example, reduce δ to respect all analytic domains and take $\|B\|_\infty, \|\omega\|_\infty < \delta/3$. This extension is an analytic identity; it is not a minimum claim for complex fields.

4.3 The exact classical Ward identity

Define the coarse left exponential differential and the coarse covariant scalar difference by

$$\mathcal{J}_B v = \int_0^1 \mathrm{Ad}(e^{itB}) v \, dt, \qquad (D_D\omega)(c) = \mathrm{Ad}_{D(c)}\, \omega(c_+) - \omega(c_-). \tag{53}$$

The operator $\mathcal{J}_B$ is invertible on the fixed small complex ball. The ordinary coarse-potential velocity under $g(t) = e^{it\omega}$ is

$$\left.\frac{d}{dt} B^{g(t)}\right|_{t=0} = -\mathcal{J}_B^{-1} D_D\omega. \tag{54}$$

The coarse spacing is one, so there is no additional length factor. Differentiate (52) and use (42):

$$0 = -\langle m(B), \mathcal{J}_B^{-1} D_D\omega\rangle_1 = -\langle D_D^T \mathcal{J}_B^{-T} m(B), \omega\rangle_1.$$

The site field ω is arbitrary. Hence

$$\boxed{D_D^T \mathcal{J}_B^{-T} m(B) = 0.} \tag{55}$$

The real identity extends holomorphically. Thus the left-coordinate classical coarse current is $j_{\mathrm{cl}}(B) = \mathcal{J}_B^{-T} m(B)$, not generally $m(B)$. No quantum determinant or quantum Ward assertion is involved.

4.4 A local classical Hessian and an auxiliary positive inverse

The weighted multiplier response (23), together with the envelope identity, gives

$$Q_{\mathrm{cl}}(B) := D^2V(B) = Dm(B), \qquad \|[Q_{\mathrm{cl}}(B) - q]c\|_{\infty,w} \le C_\kappa b \, \|c\|_{\infty,w} \tag{56}$$

for every admissible weight in (1). As the Hessian of a holomorphic scalar function, $Q_{\mathrm{cl}}(B)$ is symmetric for the physical bilinear transpose. It differentiates in the ordinary coarse potential B. A change to a based left-exponential fluctuation chart requires the coordinate Jacobians and its background-current second-derivative term; that differently based Hessian is not being identified with $Q_{\mathrm{cl}}(B)$ here.

Use a fixed trace-orthonormal color and orientation basis in each coarse cell. For a block kernel T define the weighted Schur norm

$$\|T\|_\kappa = \max\left\{ \sup_n \sum_m e^{\kappa d(n,m)} \, \|T_{nm}\| \, , \, \sup_m \sum_n e^{\kappa d(n,m)} \, \|T_{nm}\| \right\}. \tag{57}$$

Take $w_n(m) = e^{-\kappa d(n,m)}$ in (56). In each scalar output row, choose input phases to match its matrix entries. This bounds its absolute row sum with the factor $e^{\kappa d(n,m)}$. Summing the fixed number of component rows converts this to matrix-block norms; transpose symmetry gives the column bound. Consequently

$$\|Q_{\mathrm{cl}}(B) - q\|_\kappa \le C'_\kappa b. \tag{58}$$

The component factor depends only on dimension and color, not on refinement or volume. These finite-regulator row tests are bounded directions even for decaying weights.

Let a lie in a fixed interval $0 < a_- \le a \le a_+$. The flat baseline estimates give, after fixing a sufficiently small common κ,

$$\left\| (q + aI)^{-1} \right\|_\kappa \le r_\kappa \tag{59}$$

uniformly. Set $P_{\mathrm{cl}}(B) = Q_{\mathrm{cl}}(B) + aI$. For

$$C'_\kappa b \le a_-/2, \qquad r_\kappa C'_\kappa b \le 1/2, \tag{60}$$

the real positive baseline $q \ge 0$ and the Schur operator bound imply

$$\mathrm{Re}\langle x, P_{\mathrm{cl}}(B)x\rangle_{\mathrm{Herm}} \ge \frac{a_-}{2} \, \|x\|_2^2 \tag{61}$$

even at complex B. The numerical-range pairing here is Hermitian; it is distinct from the analytic bilinear transpose. The Schur norm is a matrix algebra norm, so the Neumann series also gives

$$\begin{aligned} \left\| P_{\mathrm{cl}}(B)^{-1} \right\|_\kappa &\le 2r_\kappa, \\ \left\| P_{\mathrm{cl}}(B)^{-1} - (q + aI)^{-1} \right\|_\kappa &\le 2r_\kappa^2 C'_\kappa b. \end{aligned} \tag{62}$$

This inverse is holomorphic and exponentially local in coarse-cell distance, with uniform constants.

The positive a is an auxiliary penalty added to a coarse classical Hessian. This is not a physical Yang–Mills mass gap or the precision of an integrated quantum effective action. Spatial kernel decay does not assert strict dependence on bounded background supports. Fluctuation determinants, Jacobian factors, quantum interactions and their iteration remain separate tasks. No finite total action is assigned here to arbitrary bounded fields on the infinite lattice.

5 Size of the flat constrained Gaussian fluctuations

There is a quantitative reason why the preceding smooth-background ball cannot itself be used as a uniform domain for integrating all fine Gaussian fluctuations. We prove it already at the flat reference, in four dimensions. Let $n_f = M^4 n_c$ be the numbers of fine and coarse sites of a compatible periodic regulator, where $M = L^j \geq L \geq 3$ and $\eta = M^{-1}$. Work in one fixed real trace-orthonormal color component; the flat operator acts identically on every color. Set

$$X = \ker Q \cap \ker(RD^T), \qquad K = d^T d.$$

The positive restricted Maxwell form defines a centered real Gaussian $Z \in X$ with density proportional to $\exp[-\langle Z, KZ\rangle_\eta/2]$ relative to the physical Euclidean measure on X. Its covariance operator under the physical pairing is the constrained inverse $\mathcal{C}$ used above.

Lemma 5.1. *For each real color component,*

$$\dim X = 3(n_f - n_c), \qquad \mathrm{Tr}(d\mathcal{C}d^T) = \dim X.$$

Proof. The map Q on bonds is onto because $QH = I$, hence $\dim\ker Q = 4(n_f - n_c)$. The scalar projection R has rank $n_f - n_c$: its complementary projection has the full coarse scalar rank n_c. The map RD^T from $\ker Q$ onto $\operatorname{ran} R$ is surjective. Indeed the scalar operator $T = G'R$ satisfies $Q_0 T = 0$ and $R\mathcal{L}T = R$, where $\mathcal{L} = D^T D$. For $f \in \operatorname{ran} R$, the bond field DTf has $QDTf = D_c Q_0 Tf = 0$ and $RD^T DTf = f$. Rank subtraction gives the dimension.

Let J_X embed X isometrically into the physical bond Hilbert space, and let $K_X = J_X^T K J_X$. Then $\mathcal{C} = J_X K_X^{-1} J_X^T$. Cyclicity of the finite operator trace gives $\mathrm{Tr}(d\mathcal{C}d^T) = \mathrm{Tr}(K_X K_X^{-1}) = \dim X$. □

The physical covariance identity is $\mathbb{E}\langle f, Z\rangle_\eta \langle g, Z\rangle_\eta = \langle f, \mathcal{C}g\rangle_\eta$. Thus the covariance of the actual coordinate values is $\eta^{-4}\mathcal{C}$, and that of the physical curls is $\eta^{-4} d\mathcal{C}d^T$. There are $6n_f$ positive plaquette components per color. Lemma 5.1 therefore gives the exact average

$$\boxed{\frac{1}{6n_f}\sum_p \mathbb{E}|(dZ)_p|^2 = \frac{M^4 - 1}{2}.} \tag{63}$$

In particular at least one real curl coordinate has variance at least $(M^4 - 1)/2$, irrespective of the periodic volume. This conclusion uses the physical covariance normalization; omitting η^{-4} would give a different and incorrect scale.

Fix a deterministic background A_0, a radius $r > 0$ and a coupling factor $g > 0$. The event $N_\alpha(A_0 + gZ) \leq r$ bounds every physical first difference, and hence bounds the selected curl coordinate by Cr, where the color-norm constant is fixed. It restricts a one-dimensional centered Gaussian of standard deviation at least $c\eta^{-2}$ to an interval of length at most Cr/g; the interval center may depend on A_0. The maximum of the Gaussian density therefore proves

$$\boxed{\mathbb{P}\{N_\alpha(A_0 + gZ) \leq r\} \leq \min\left(1, \frac{Cr\eta^2}{g}\right).} \tag{64}$$

The bound is uniform in the deterministic background, refinement and compatible periodic volume. Thus for any sequence $g = g_\eta$ with $g_\eta/\eta^2 \to \infty$, a fixed physical smooth ball has probability tending to zero under this constrained flat Gaussian reference.

This is an obstruction to integrating all fluctuations inside the minimizer's fixed N_α ball, not an obstruction to a Yang–Mills theory. Equation (63) shows that the dimensionless microscopic curl $\eta^2 dZ$ has bounded averaged variance. A quantum renormalization step must instead control suitable rescaled fluctuation fields and their tails while retaining smooth estimates for the background. Averaged or component variance bounds alone do not control a supremum over a growing number of variables, nor do they bound interacting polymer activities. No non-Gaussian probability estimate is inferred here.

6 The next interface to quantum integration

The classical background and its response are now controlled on a uniform homogeneous small-field domain. The minimized action has a true nonlinear local response and a coarse Hessian close to the flat Maxwell precision. Adding an auxiliary positive coarse penalty gives a controlled inverse, but that penalty is not a physical mass and this inverse is not yet the covariance produced by the complete quantum renormalization step.

Exponential response alone does not give strict background support for polymer activities. The physical Hessian, gauge-coordinate factors and multiple-insertion local decomposition require the separate constructions described below. Regions outside the chart and preservation of uniform complex-source bounds under quantum iteration remain essential.

Notebook 17 now proves a relative energy comparison for the actual first-step reduced Hessian, retaining constraint curvature, and joint Gaussian bad-block estimates on the fixed-step domain. Its homogeneous flat current-scale theorem covers all previous averaging depths. Notebook 18 handles overlapping local interaction supports and strict background locality under a small local precision decomposition. Notebook 19 supplies the actual physical pivot Hessian and local measure factors. Notebook 20 closes the bare Wilson small-field integration. Notebook 21 supplies strict support for the minimized classical action, with arbitrary fixed cardinality reserve, and extends one classical-reference integration uniformly through earlier refinement. The next-coarse support transport, inherited quantum interactions and regional domains remain separate.

The continuum obligations remain those of Notebook 01: a common nontrivial gauge-invariant observable hierarchy, Euclidean covariance and positivity in the limit, appropriate normalization and Yang–Mills identities, and the additional vacuum and mass-gap properties. The classical theorems here remove specific prerequisites; they do not establish those quantum conclusions.

References

[1] *Toward a four-dimensional Yang–Mills construction*, research Notebooks 8–15 in the present workspace, 16 October 2026. The weighted flat extension and inverse, uniform nonlinear constraint chart, Wilson force, constrained minimum and ordinary-average transfer are the prerequisites.

[2] E. Sussman and collaborators, Yang–Mills-LLM-experiment, deposited raw accounts I–III at commit `e392ef45013a581664bab76b90867170b30fe10e`. The provenance limitations of Notebook 15 remain in force.